

# Industrial Equipment Outlier Detection

Data storytelling presentation with all generated charts

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Learning Day 3 - Outlier Detection | Z-Score, Isolation Forest, LOF, ABOD, CBLOF and LoOP



# Executive Summary

The final decision and why it matters

## DATASET

**7,672 rows**

Industrial equipment sensor observations across temperature, pressure, vibration and humidity.

## METHODS COMPARED

**6 algorithms**

Z-Score, Isolation Forest, LOF, ABOD, CBLOF and LoOP.

## FINAL WINNER

**Z-Score**

Best F1-score against the available faulty label.

## F1-SCORE

**0.7504**

Best balance of precision and recall on this dataset.

## Main storyline

Faulty equipment rows were strongly linked to extreme sensor values. Because of this, the simple statistical Z-Score method outperformed more advanced model-based detectors on the available faulty label.

## Final model evidence

Z-Score: Precision 0.9914 | Recall 0.6037 | F1-score 0.7504. CBLOF and Isolation Forest were precise, but missed more actual faulty rows.

# Problem Framing and Dataset

What we are detecting and how success is measured

## Business question

Which outlier detection method is most useful for identifying faulty equipment observations from sensor data?

## Available data

Each row contains four main sensor readings and a faulty label. The faulty label was used only for evaluation, not to train unsupervised detectors.

TEMPERATURE

**sensor**

PRESSURE

**sensor**

VIBRATION

**sensor**

HUMIDITY

**sensor**

## Success metrics used

Precision: when a row is flagged, is it really faulty? | Recall: how many actual faulty rows were caught? | F1-score: balance between precision and recall.

# Method Map

Different algorithms define unusual behavior in different ways

## Z-Score

### STATISTICAL EXTREMES

Flags rows with sensor values far from the mean.

## Isolation Forest

### TREE-BASED ISOLATION

Finds rows that are easy to isolate using random splits.

## LOF

### LOCAL DENSITY

Finds rows that look unusual compared with nearby rows.

## ABOD

### ANGLE GEOMETRY

Finds unusual geometric positions using angle-based logic.

## CBLOF

### CLUSTER DISTANCE

Finds rows far from or poorly represented by clusters.

## LoOP

### LOCAL PROBABILITY

Uses probability-style local anomaly scores.

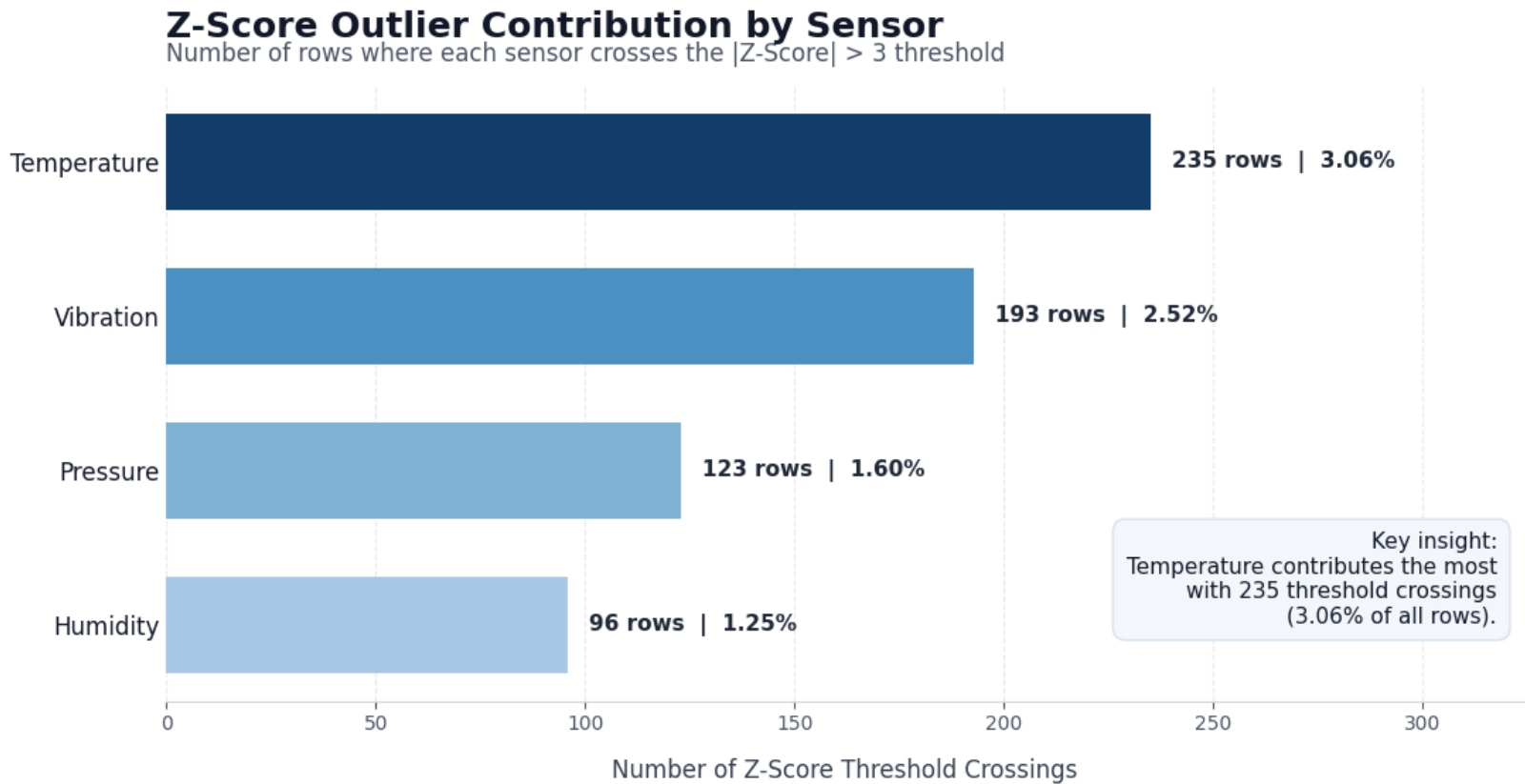
## PART 1

# Statistical Outlier Detection

We started with Z-Score to identify rows with extreme individual sensor readings. This establishes a simple, interpretable baseline.

# Z-Score Outlier Contribution by Sensor

Z-Score baseline



Note: A single row can be counted under multiple sensors if multiple sensor values exceed the Z-Score threshold.

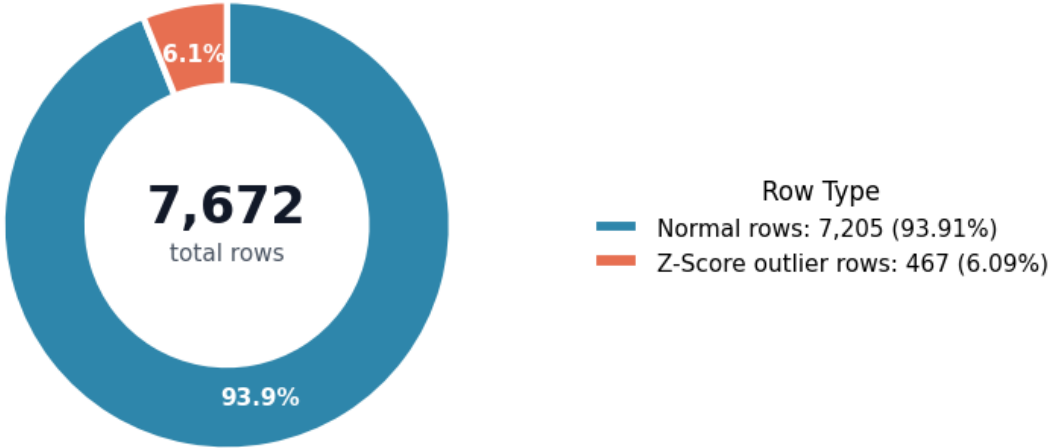
#### Key takeaway

Temperature contributes the largest number of Z-Score outliers, followed by vibration. This immediately tells us which sensors drive most extreme values.

# Normal vs Z-Score Outlier Rows

Z-Score baseline

Normal Rows vs Z-Score Outlier Rows



Key takeaway

Z-Score identified 467 outlier rows, about 6.09% of all observations. This is a compact but meaningful anomalous segment.

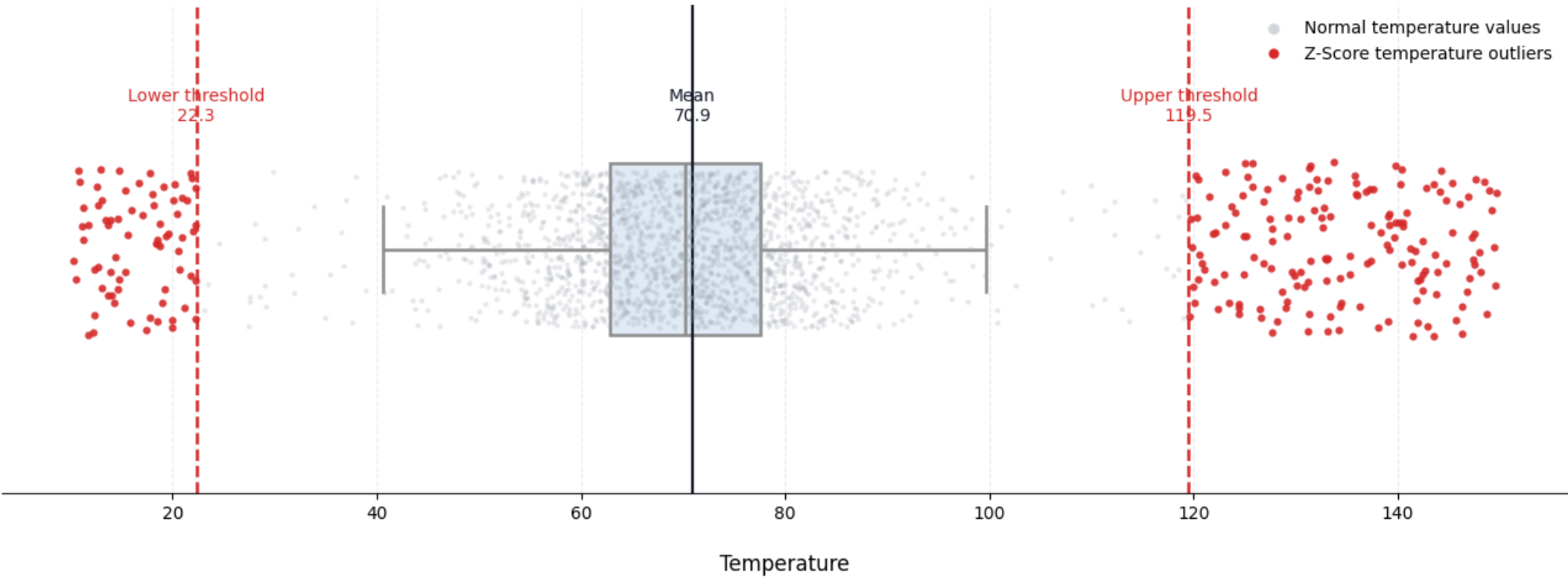
# Temperature Distribution with Z-Score Outliers

Z-Score baseline

Chart 3/20

## Temperature Distribution with Z-Score Outliers Highlighted

Red points show temperature values where  $|\text{temperature Z-Score}| > 3$

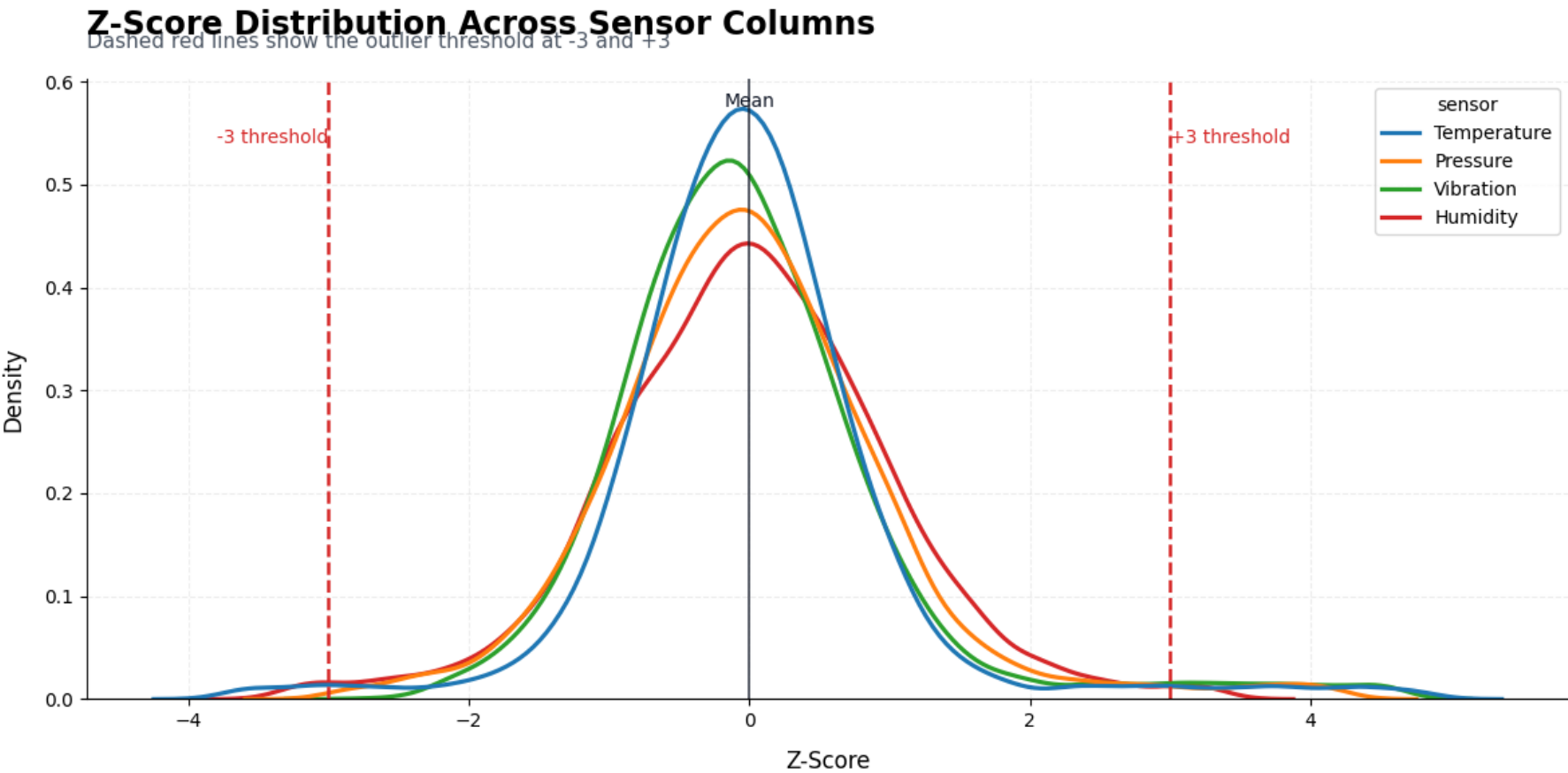


### Key takeaway

The highlighted rows sit in extreme temperature regions, which supports the idea that fault-like behavior is often tied to sensor extremes.

# Z-Score Distribution Across Sensors

Z-Score baseline



#### Key takeaway

The sensor-wise Z-Score distributions show how each variable contributes to extreme readings and where the threshold cuts the data.

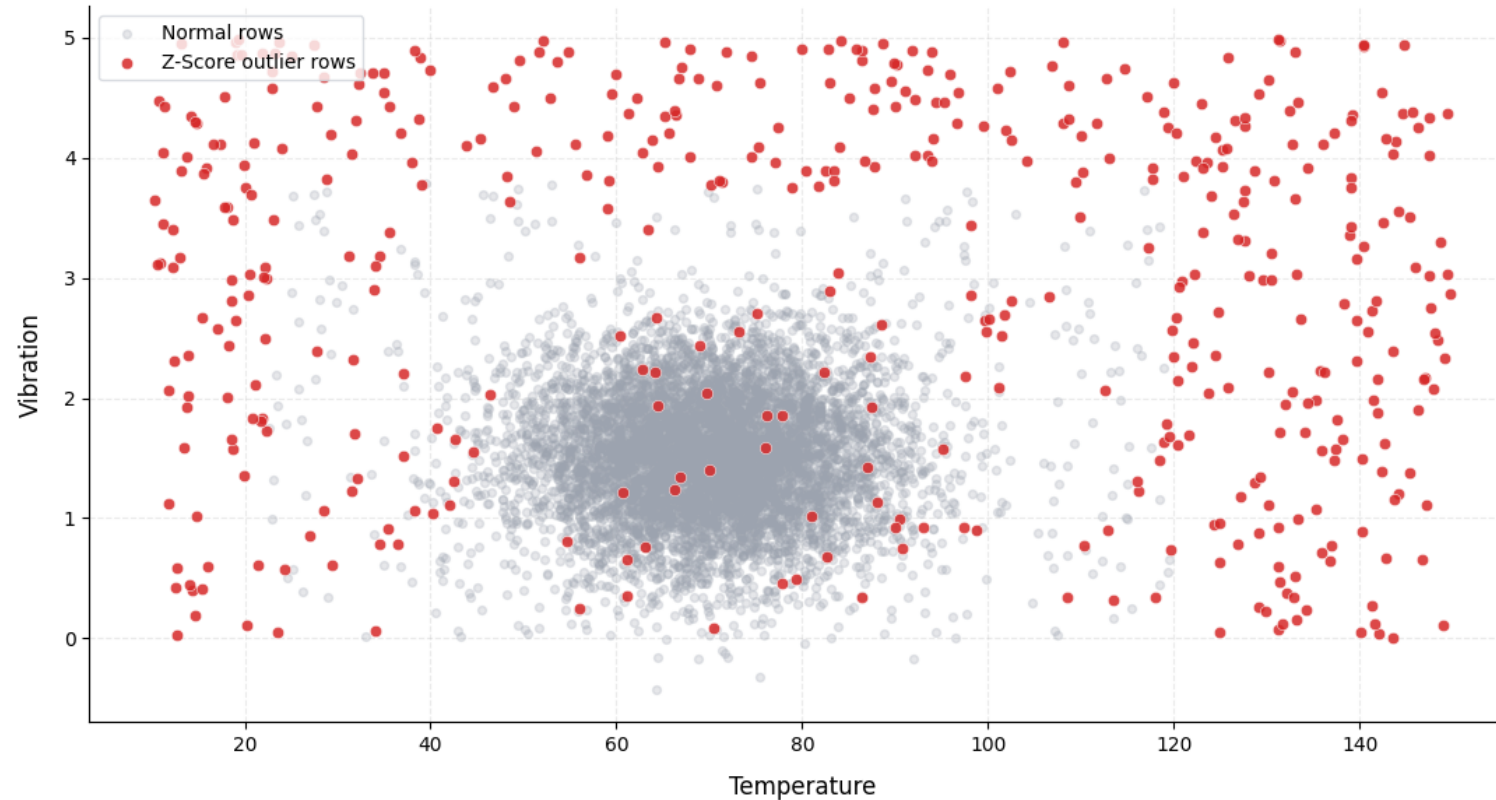
# Temperature vs Vibration Outlier View

Z-Score baseline

Chart 5/20

## Temperature vs Vibration with Z-Score Outliers Highlighted

Outliers are rows where at least one sensor has  $|Z\text{-Score}| > 3$



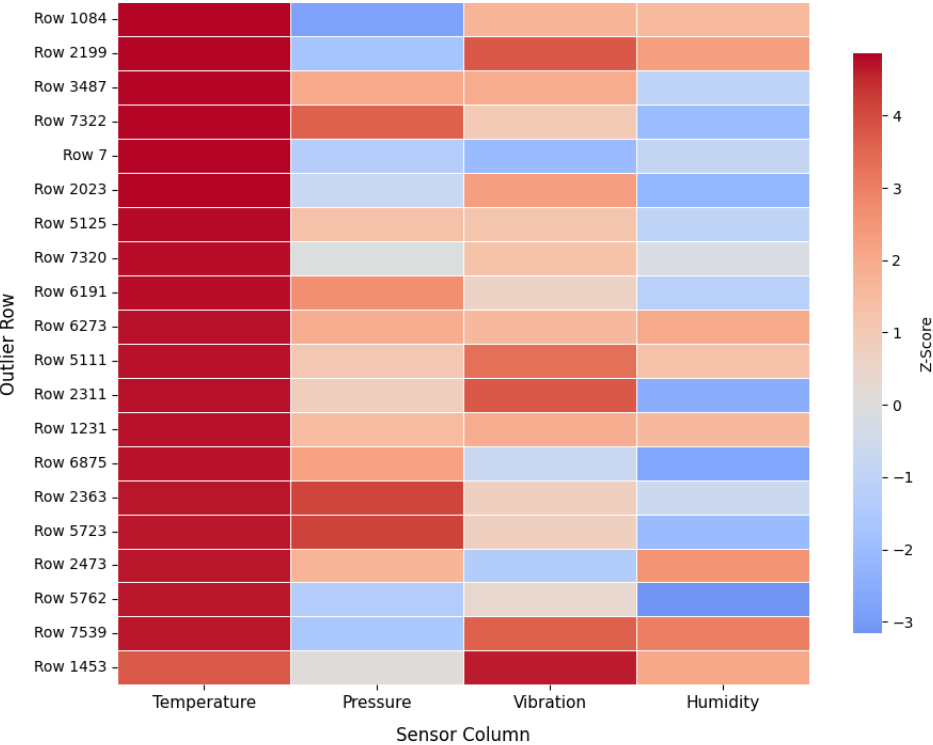
### Key takeaway

The scatter view shows how Z-Score outliers separate from the main cloud, especially when temperature and vibration are considered together.

# Top Z-Score Outlier Rows by Sensor

Z-Score baseline

**Top Z-Score Outlier Rows by Sensor**  
Each row shows one strong outlier; color intensity shows how abnormal each sensor value is.



## Key takeaway

The heatmap reveals that many top outlier rows are abnormal across multiple sensors, not just a single isolated spike.

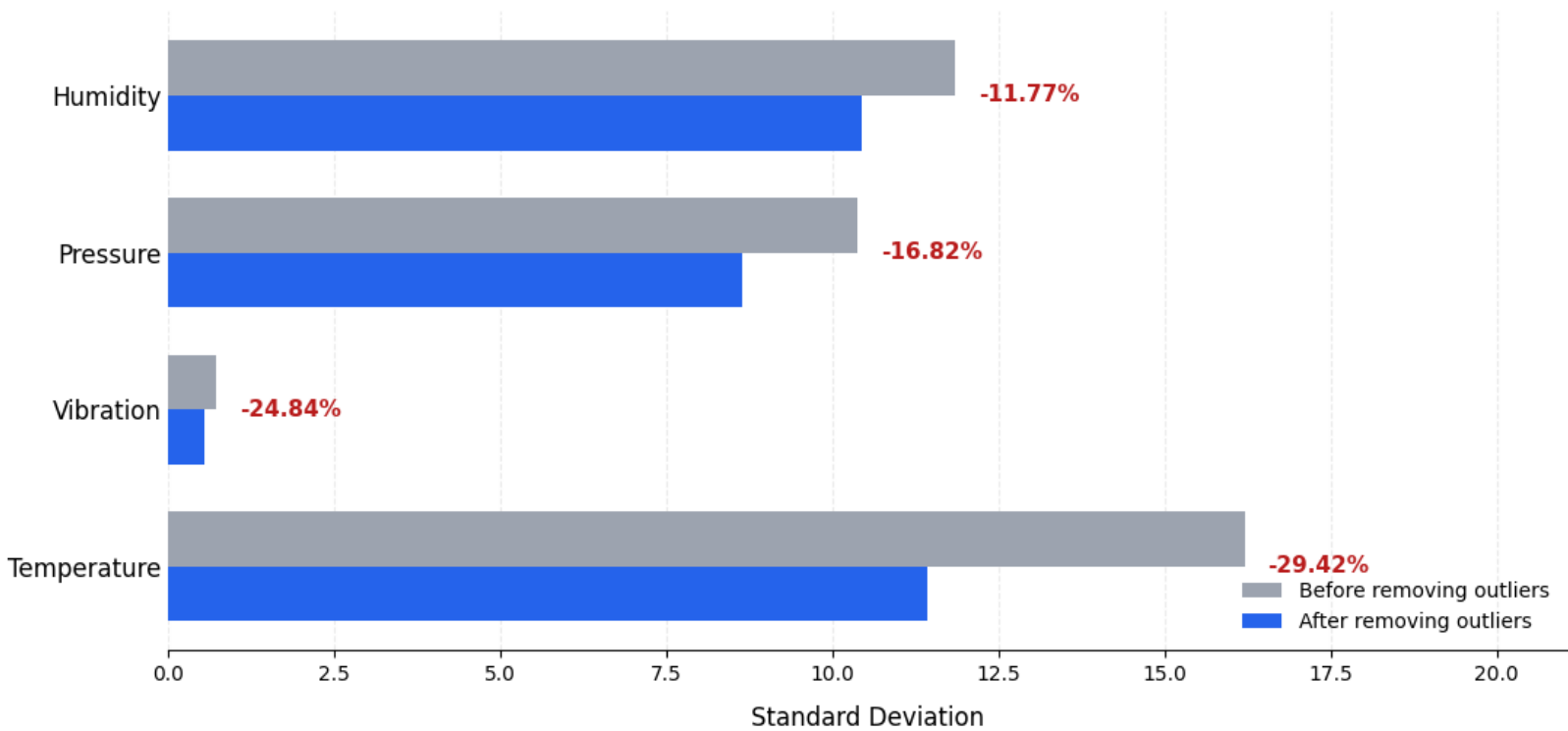
# Impact of Removing Z-Score Outliers

Z-Score baseline

Chart 7/20

## Standard Deviation Reduced After Removing Z-Score Outliers

Negative percentages show how much sensor variability decreased after outlier removal



Note: Standard deviation measures spread. Lower values after removal indicate a more stable sensor distribution.

### Key takeaway

After removing Z-Score outliers, standard deviation drops across sensors. This confirms that flagged rows were driving excess variability.

## PART 2

# Model-Based Detectors

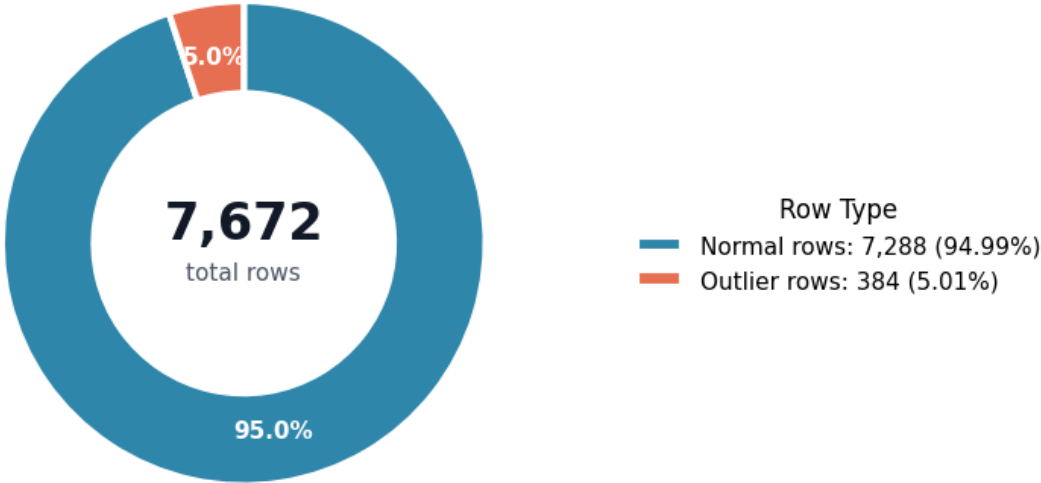
Next, we compared Isolation Forest and LOF. Both detected 384 rows, but they did not agree on which rows were outliers.

# Isolation Forest Detection Summary

Isolation Forest and LOF

Chart 8/20

## Isolation Forest Detection Summary



### Key takeaway

Isolation Forest detected 384 rows, exactly 5.01% of the dataset. This follows the configured contamination level.

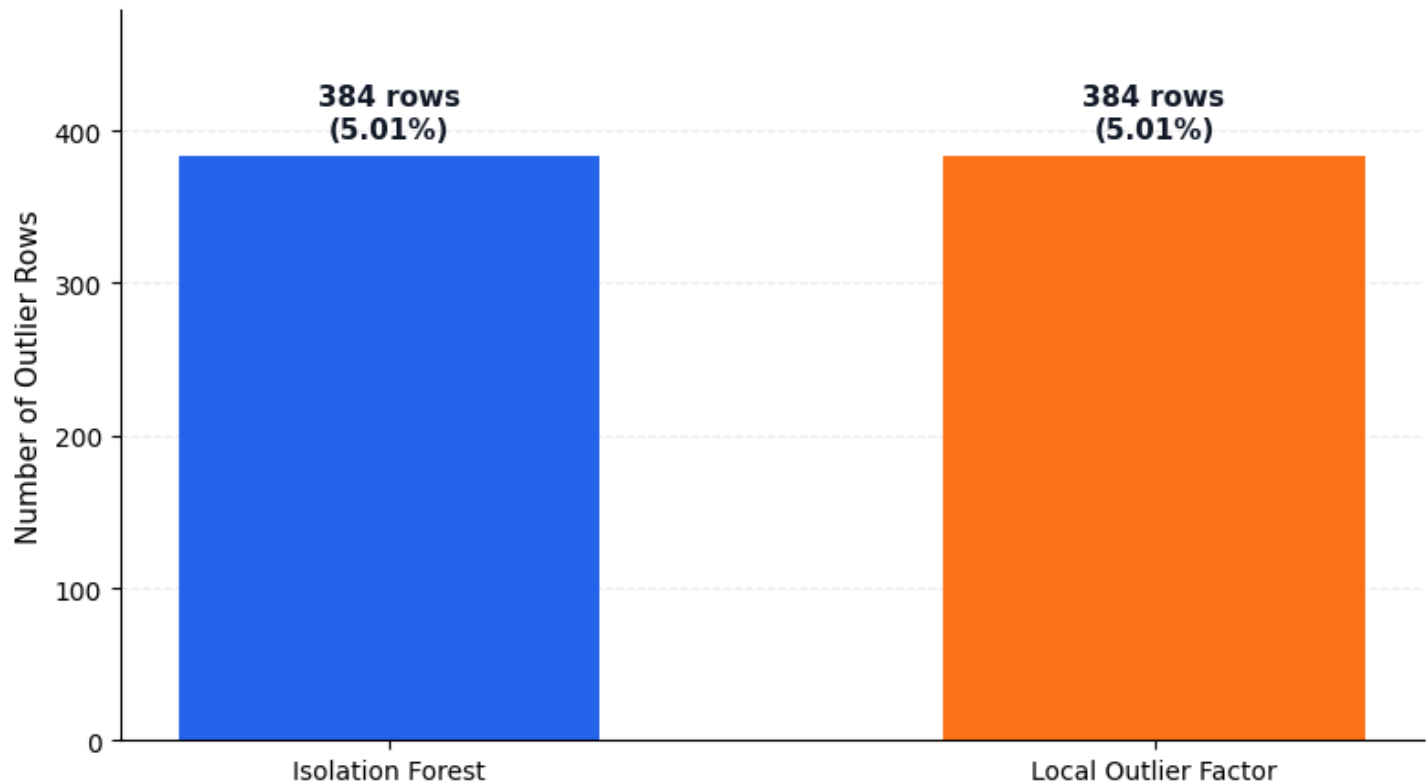
# Isolation Forest vs LOF: Count Comparison

Isolation Forest and LOF

Chart 9/20

## Outlier Count Comparison: Isolation Forest vs LOF

Both models used contamination = 0.05, so both flag approximately 5% of rows



### Key takeaway

Isolation Forest and LOF flagged the same number of rows, but this alone does not mean they found the same observations.

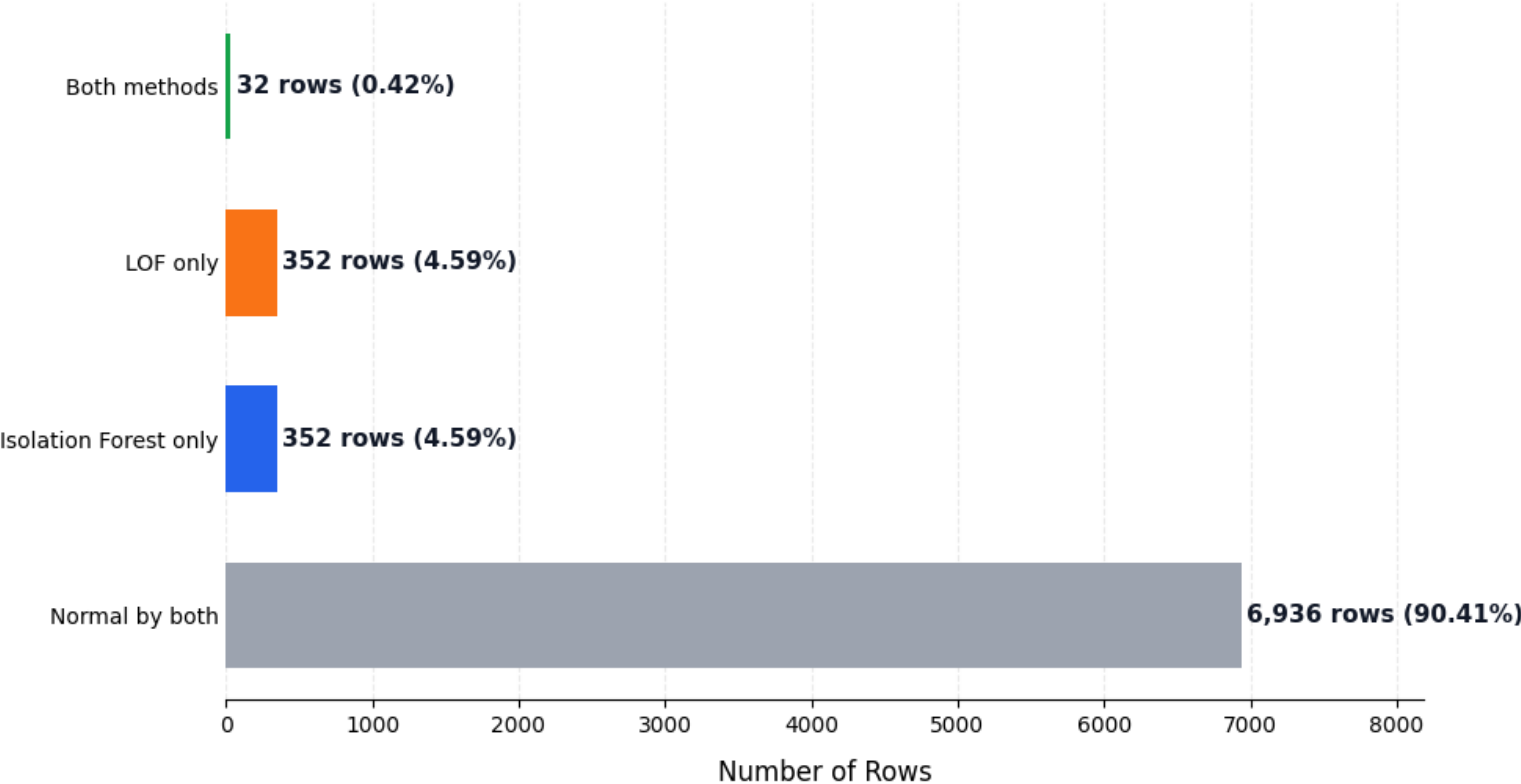
# Agreement Between Isolation Forest and LOF

Isolation Forest and LOF

Chart 10/20

## Agreement Between Isolation Forest and LOF

Only 32 rows were marked as outliers by both methods; Jaccard = 0.0435



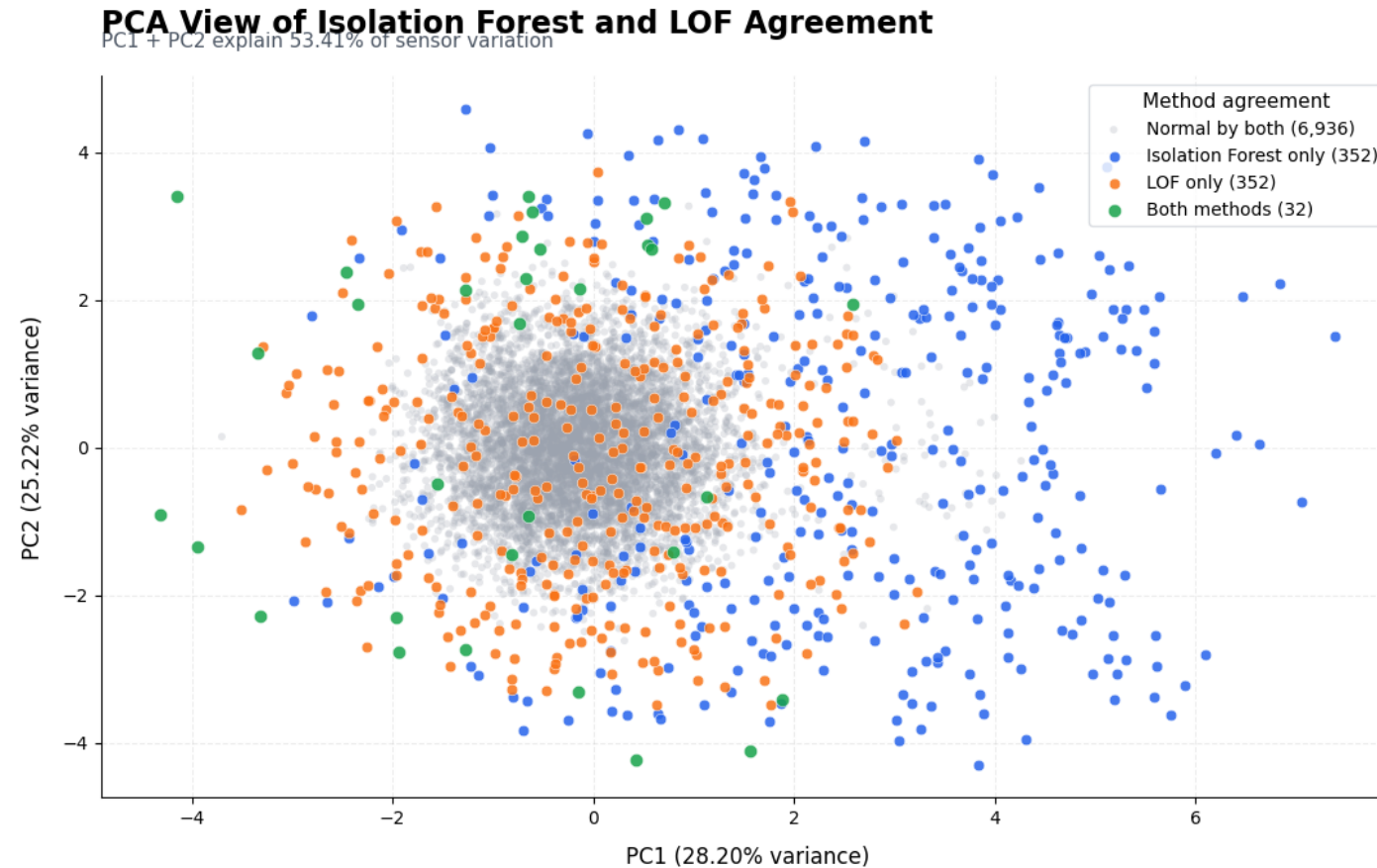
### Key takeaway

Only 32 rows were shared by both methods. The Jaccard score was 0.0435, meaning agreement was very low.

# PCA View of Isolation Forest and LOF Agreement

Isolation Forest and LOF

Chart 11/20



## Key takeaway

PCA shows that the two detectors highlight different regions of the sensor space. PC1 and PC2 explain 53.41% of sensor variation.

## PART 3

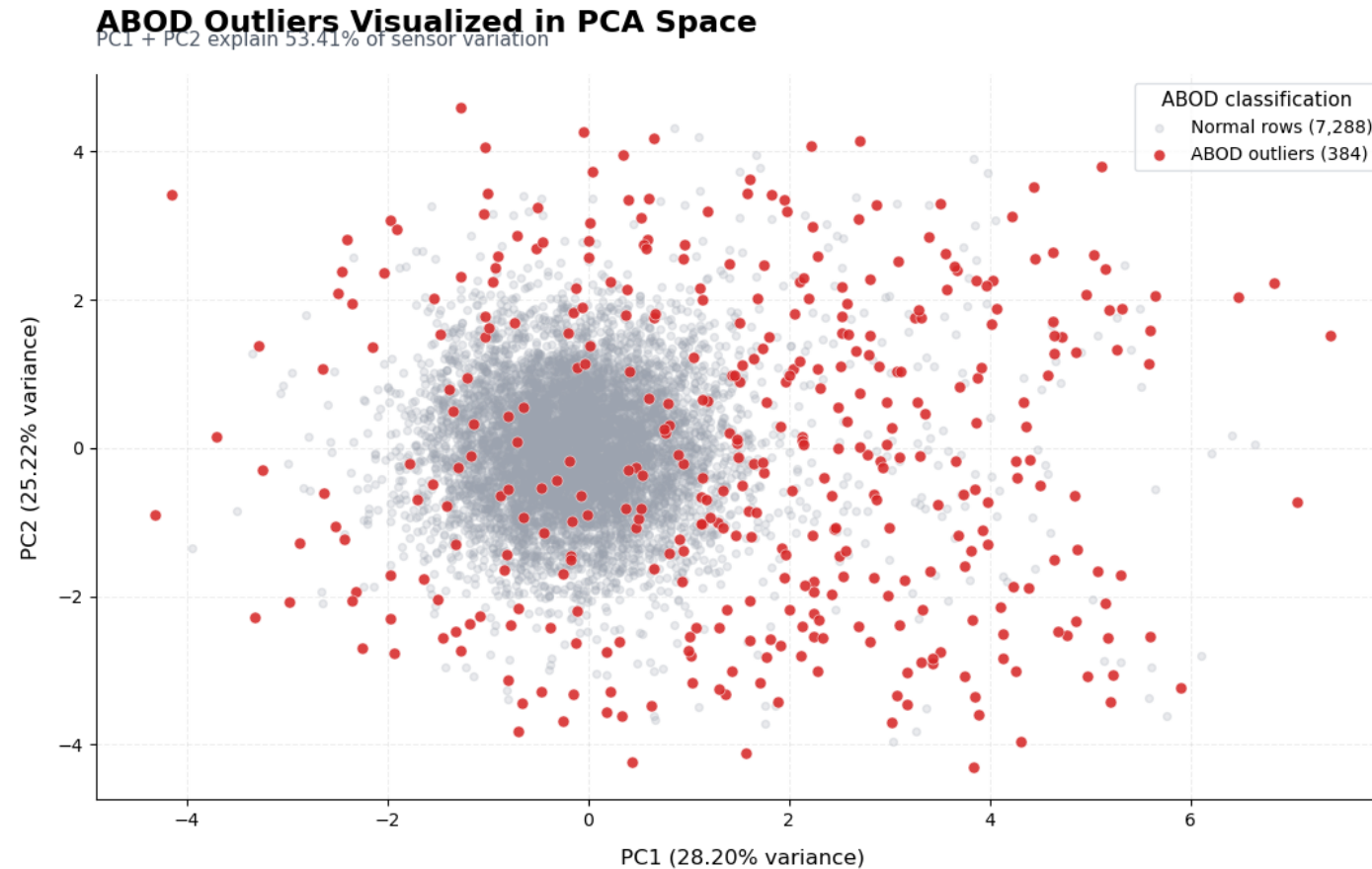
# Extended Outlier Methods

We added ABOD, CBLOF and LoOP to understand whether more specialized anomaly definitions improve detection.

# ABOD Outliers in PCA Space

Extended methods

Chart 12/20



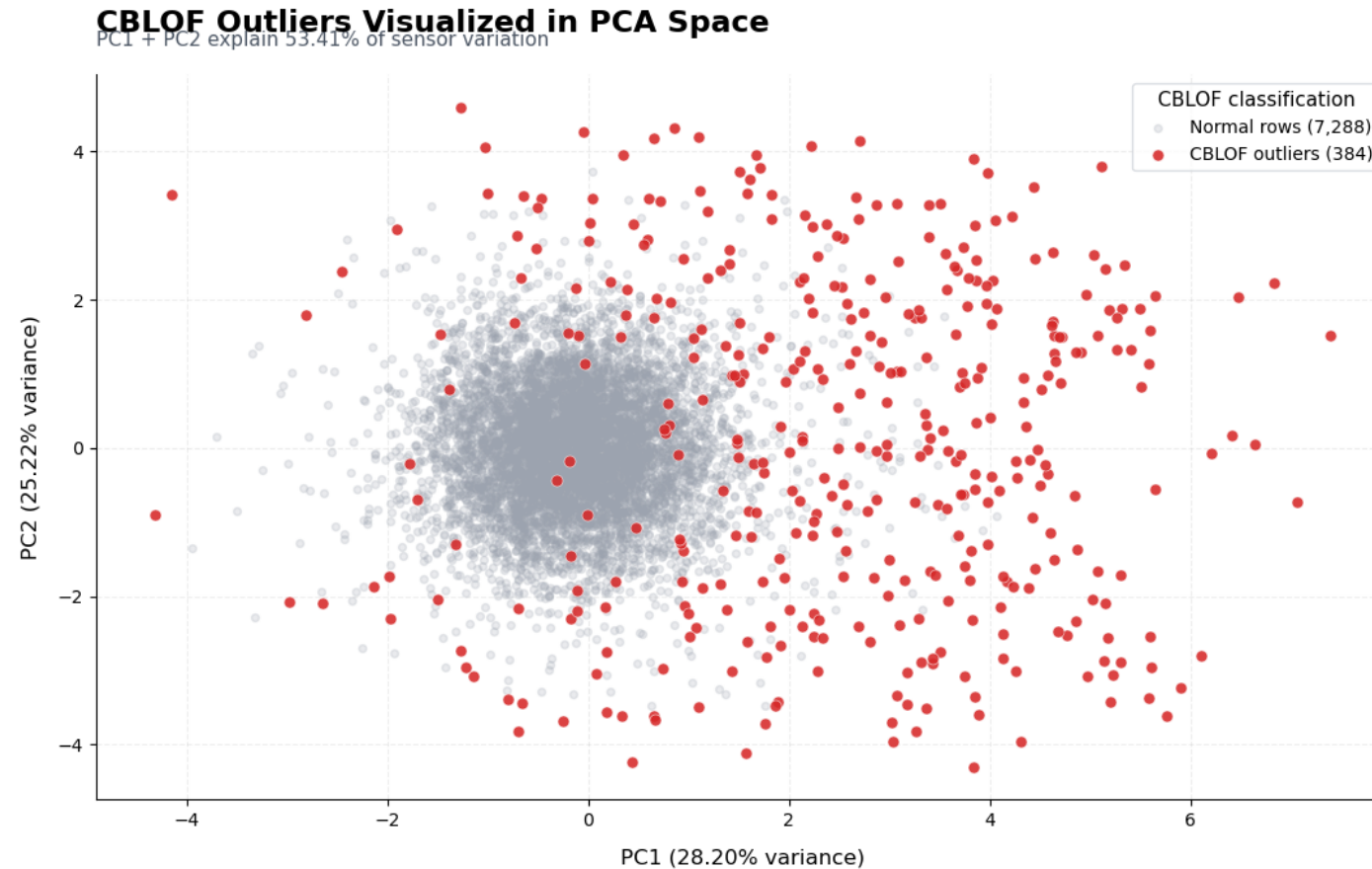
## Key takeaway

ABOD identifies angle-based anomalies across the PCA map, showing a geometric view of unusual observations.

# CBLOF Outliers in PCA Space

Extended methods

Chart 13/20



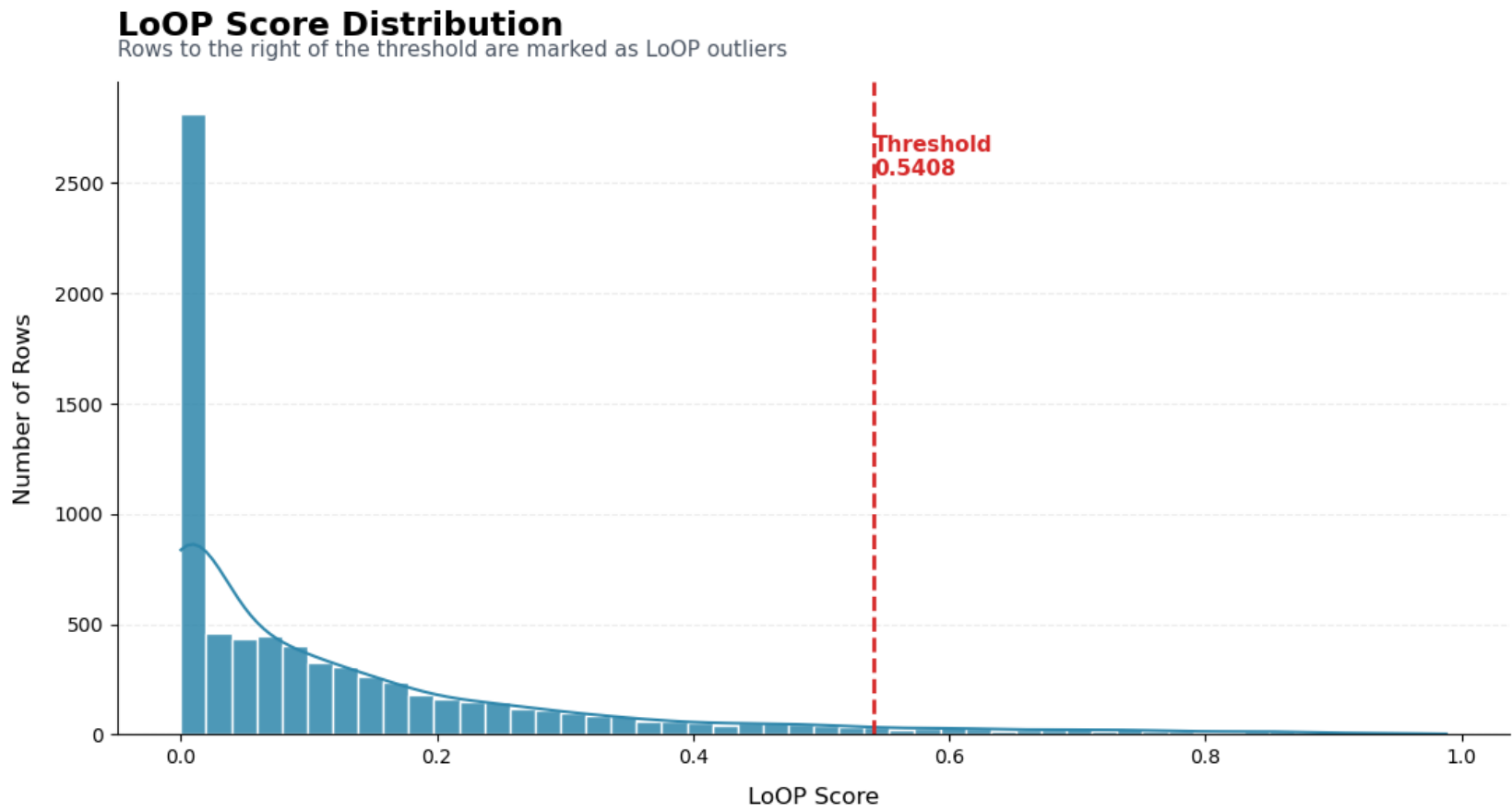
## Key takeaway

CBLOF highlights cluster-distance anomalies, with many flagged rows appearing away from the dense center of normal points.

# LoOP Score Distribution

Extended methods

Chart 14/20



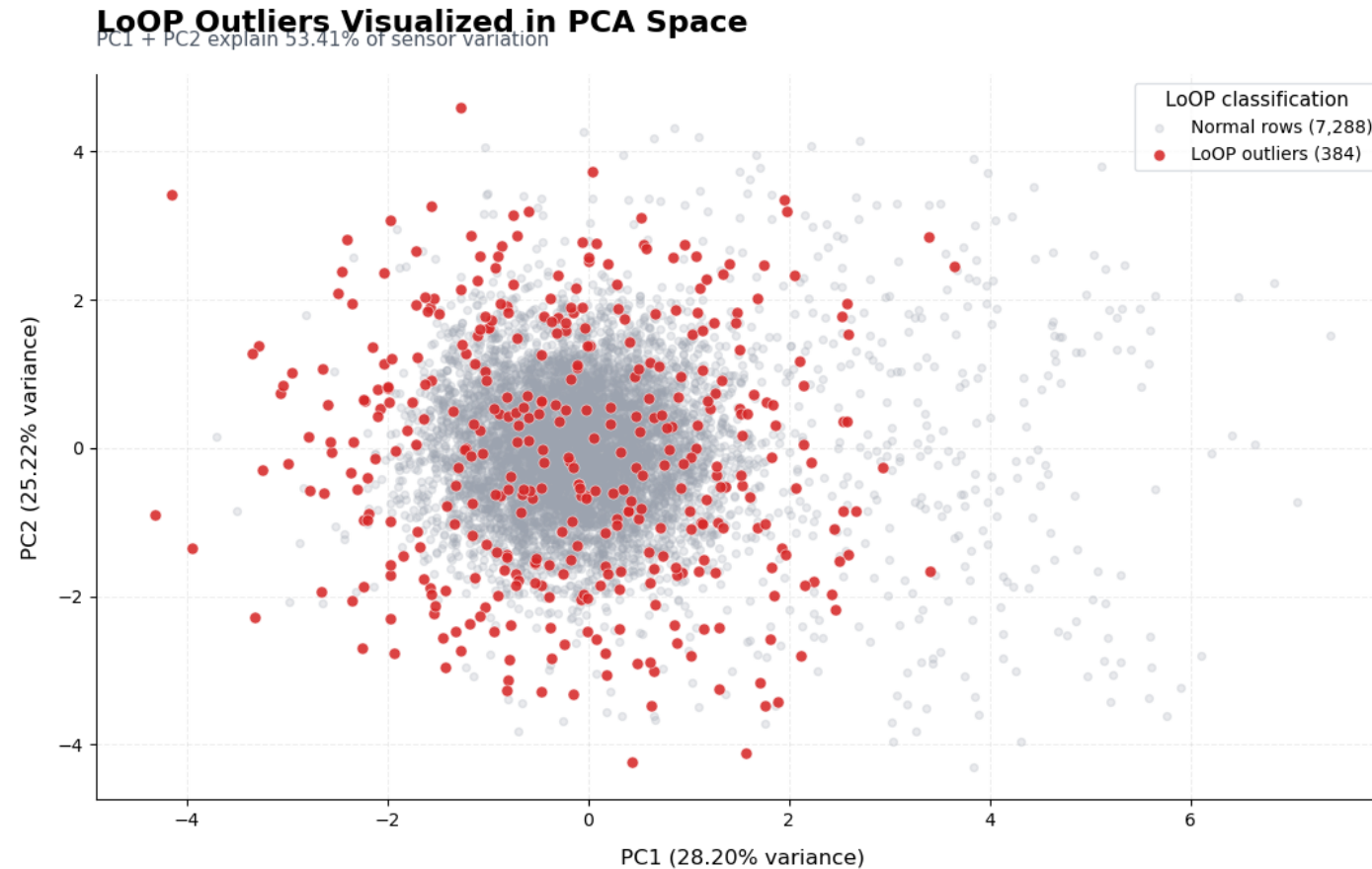
**Key takeaway**

LoOP uses a probability-style anomaly score. Rows above the threshold of 0.5408 were flagged as outliers.

# LoOP Outliers in PCA Space

Extended methods

Chart 15/20



## Key takeaway

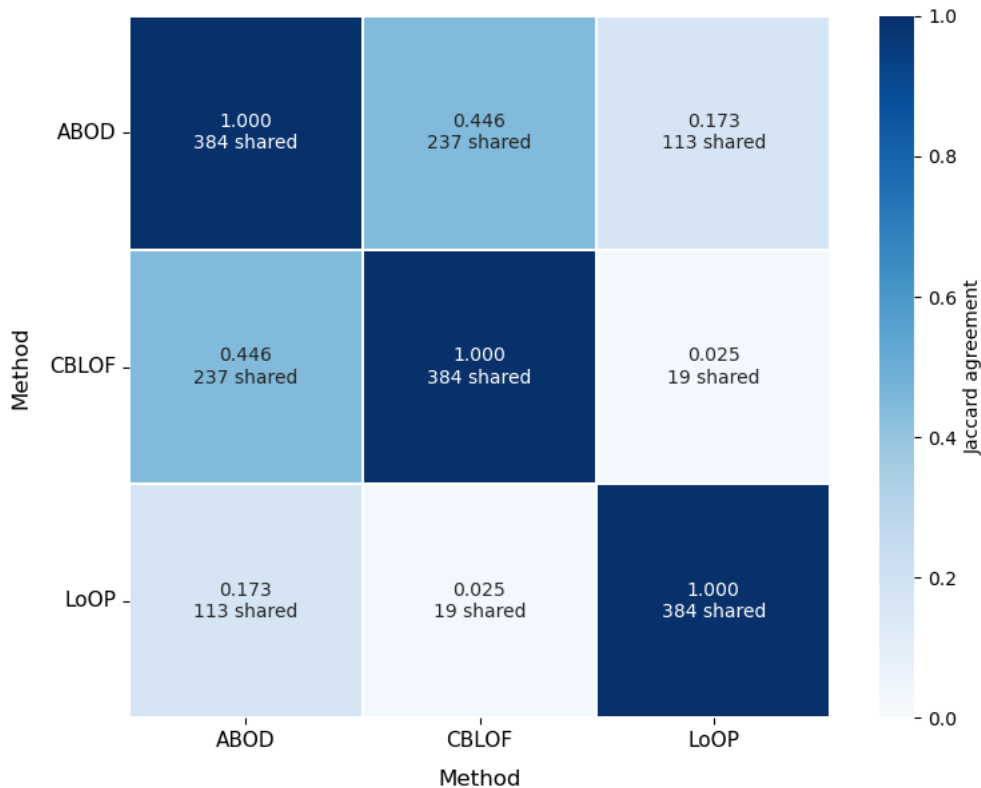
LoOP focuses on local probability anomalies and captures a different pattern from CBLOF and ABOD.

# Pairwise Agreement Between Extended Methods

Extended methods

Chart 16/20

Pairwise Agreement Between Extended Outlier Methods



Key takeaway

ABOD and CBLOF agree most strongly. LoOP has much lower overlap, meaning it captures a different anomaly definition.

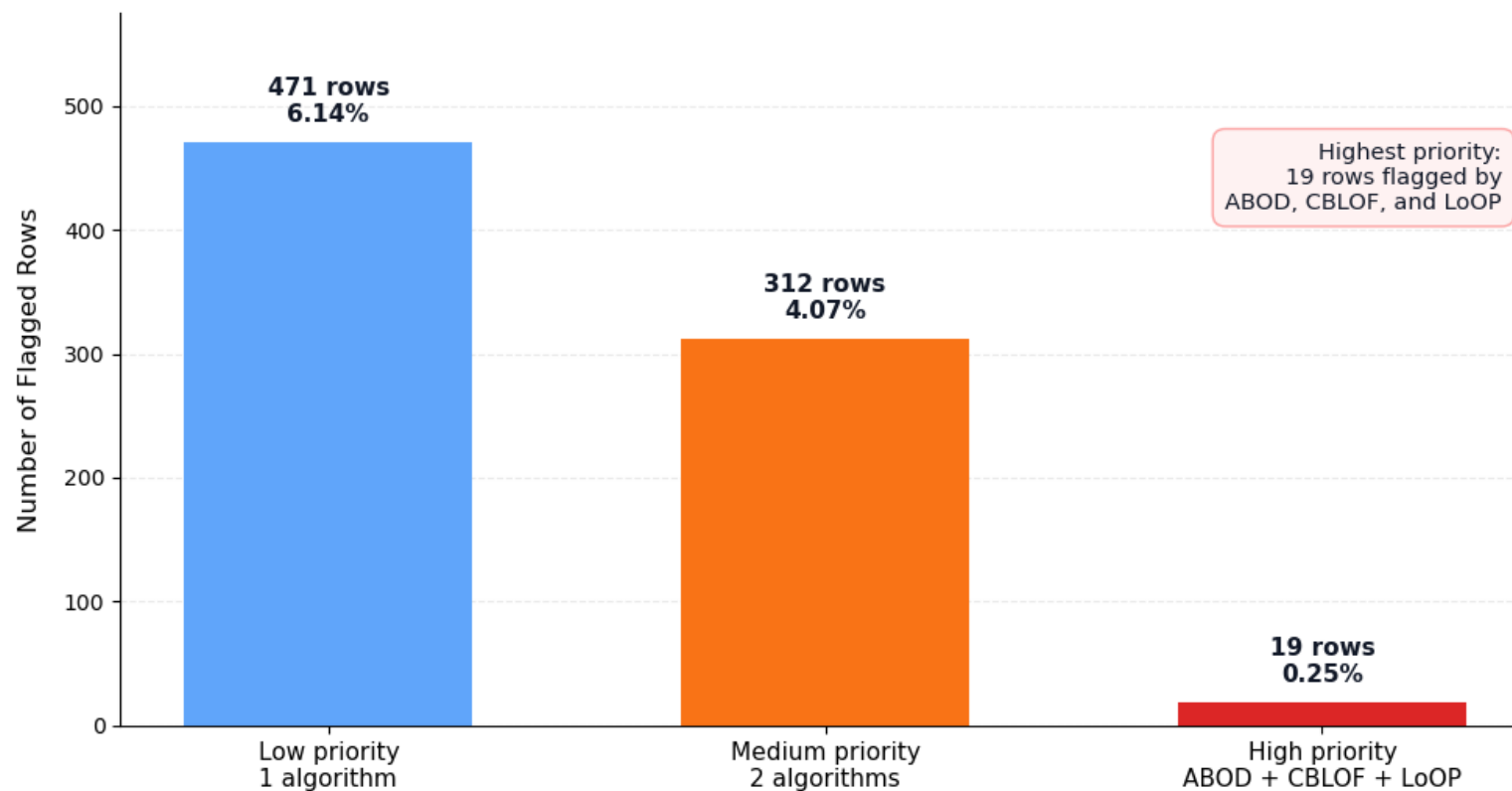
# Priority Based on ABOD, CBLOF and LoOP Agreement

Extended methods

Chart 17/20

## Anomaly Priority Based on ABOD, CBLOF, and LoOP Agreement

Only rows flagged by at least one algorithm are shown



### Key takeaway

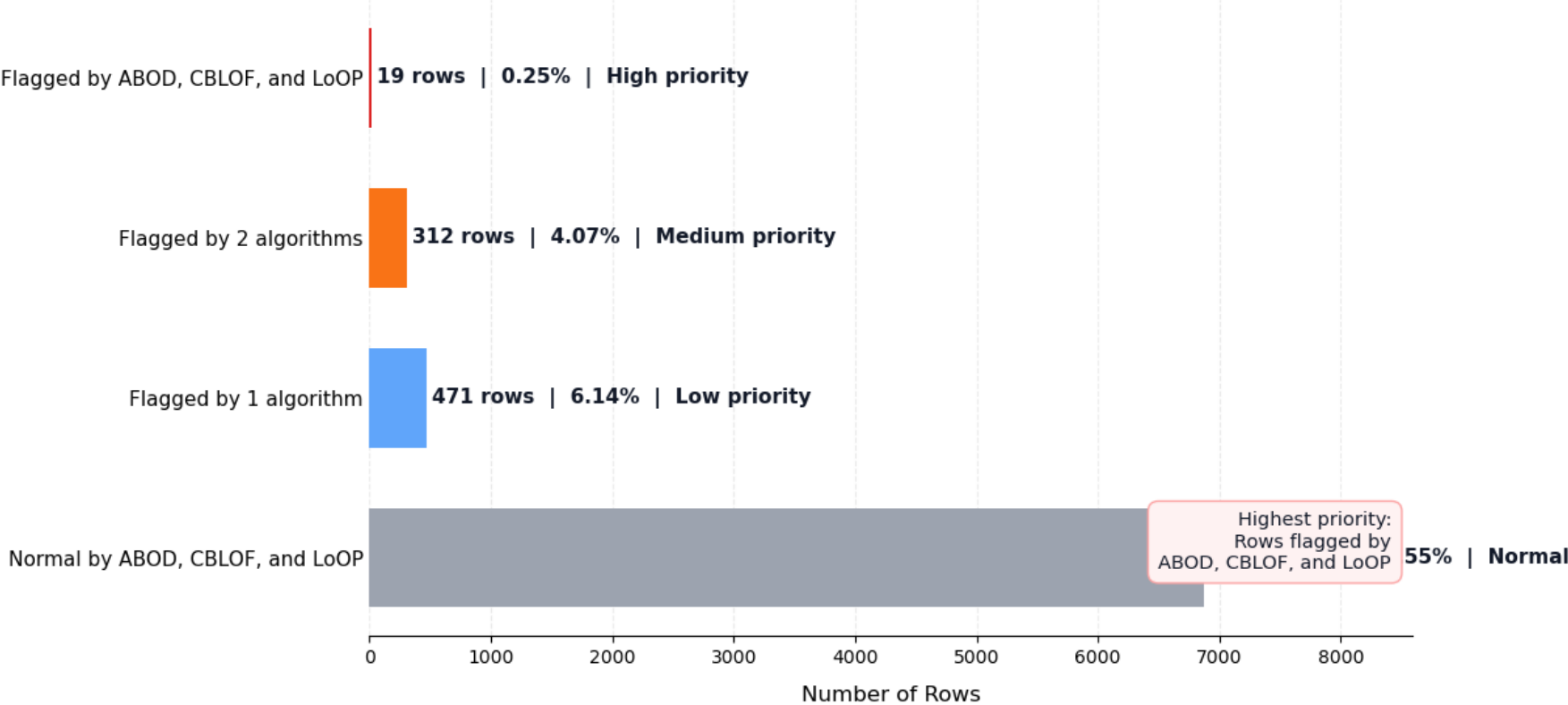
The highest-priority group contains rows flagged by all three extended methods. These are strong candidates for manual investigation.

# Agreement Across ABOD, CBLOF and LoOP

Extended methods

Chart 18/20

**Agreement Across ABOD, CBLOF, and LoOP**  
Rows flagged by more algorithms are treated as stronger anomaly candidates



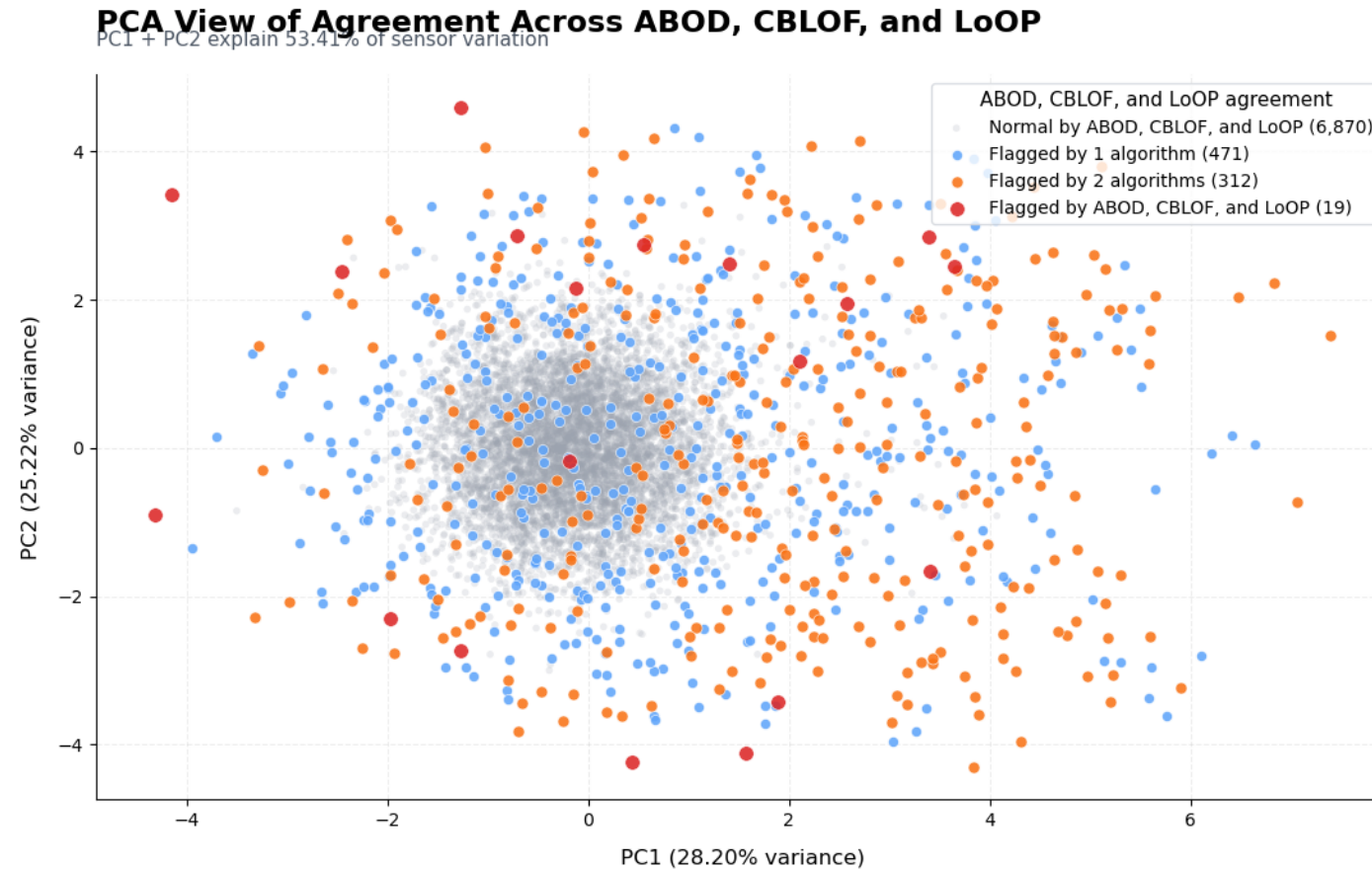
**Key takeaway**

Most rows were normal by all three methods. Rows flagged by multiple methods form a smaller, higher-confidence anomaly set.

# PCA View of Three-Method Agreement

Extended methods

Chart 19/20



## Key takeaway

The PCA map shows how consensus anomalies are distributed across the sensor space, separating low, medium and high-priority candidates.

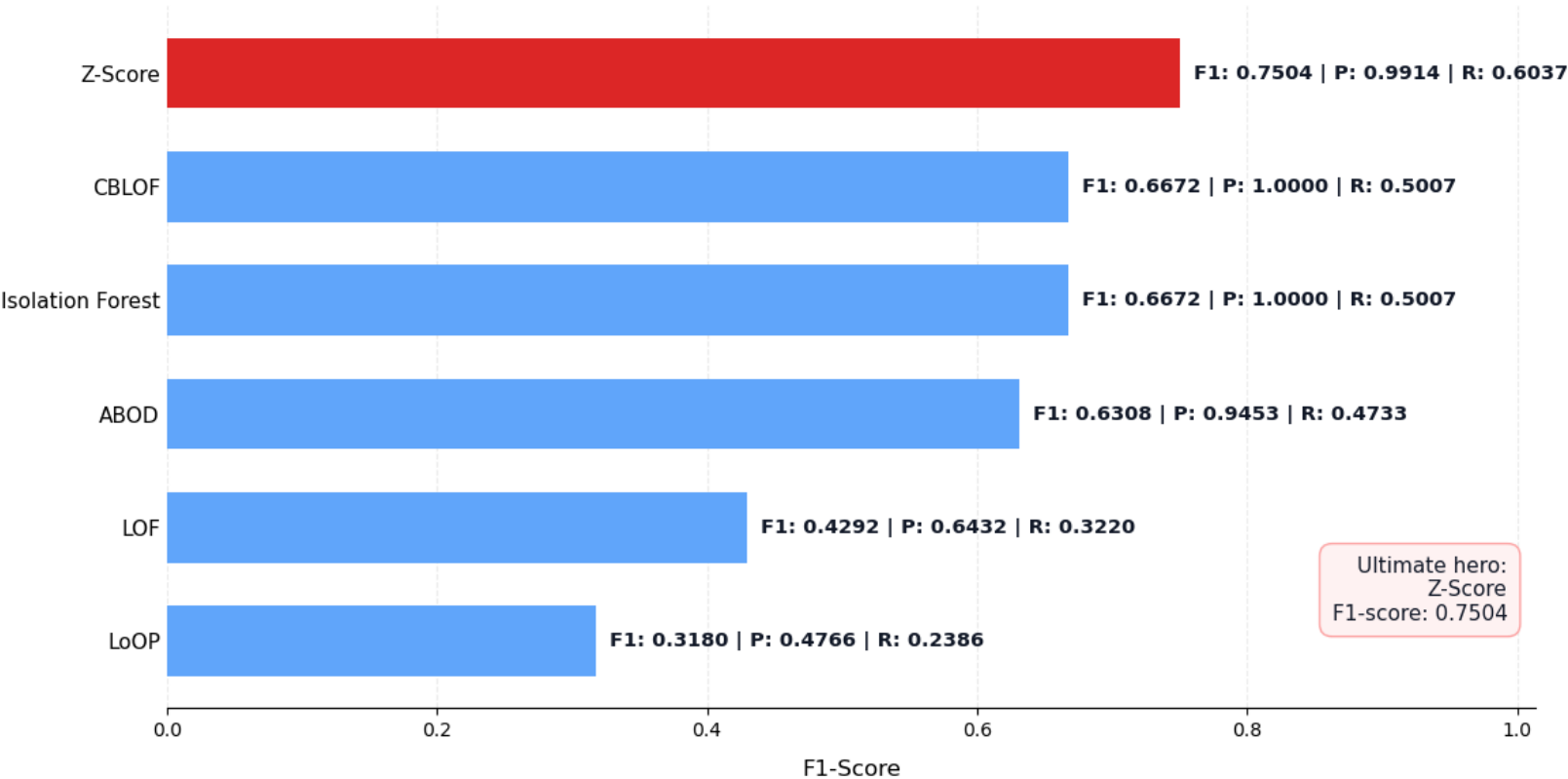
# Master Comparison: Which Algorithm Performs Best?

Final comparison

Chart 20/20

## Master Comparison: Which Outlier Algorithm Performs Best?

Algorithms compared using F1-score against the available faulty label



### Key takeaway

Z-Score achieved the highest F1-score. In this dataset, simple statistical extremes aligned best with the available faulty label.

# Final Leaderboard

Performance against the available faulty label

| Algorithm        | Precision | Recall | F1-score | Interpretation                                |
|------------------|-----------|--------|----------|-----------------------------------------------|
| Z-Score          | 0.9914    | 0.6037 | 0.7504   | Best overall balance; final winner.           |
| CBLOF            | 1.0000    | 0.5007 | 0.6672   | Perfect precision but lower recall.           |
| Isolation Forest | 1.0000    | 0.5007 | 0.6672   | Same F1 as CBLOF.                             |
| ABOD             | 0.9453    | 0.4733 | 0.6308   | Strong but below top methods.                 |
| LOF              | 0.6432    | 0.3220 | 0.4292   | Local density less aligned with faulty label. |
| LoOP             | 0.4766    | 0.2386 | 0.3180   | Lowest match to faulty label.                 |

## Recommendation

Use Z-Score as the primary detector for this dataset because it is interpretable, simple to explain, and achieved the strongest F1-score. Use agreement from ABOD/CBLOF/LoOP as a secondary prioritization layer for manual inspection.

# Final Data Story

The analysis tested increasingly complex outlier detection methods. The strongest result came from the simplest method: Z-Score. This tells us that in this industrial equipment dataset, faulty rows are closely connected to statistically extreme sensor values.

WINNER

## Z-Score

Highest F1-score: 0.7504

BEST PRECISION

## CBLOF / Isolation Forest

Both reached 1.0000 precision

HIGH-PRIORITY REVIEW

## 19 rows

Flagged by ABOD + CBLOF + LoOP